

Trigonometric Functions & the Unit Circle

ANSWER KEY



Radians, degrees, and the unit circle

- 1 Convert 150° to radians, and convert $\frac{5\pi}{4}$ radians to degrees.
 $150^\circ \cdot \frac{\pi}{180^\circ} = \frac{5\pi}{6}$ radians. $\frac{5\pi}{4} \cdot \frac{180^\circ}{\pi} = 225^\circ$.
- 2 Find the exact coordinates of the point on the unit circle at angle $\theta = \frac{2\pi}{3}$.
 $\frac{2\pi}{3} = 120^\circ$, a reference angle of 60° in Quadrant II. Coordinates: $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

Exact values from the unit circle

- 3 Find the exact values of $\sin\left(\frac{7\pi}{6}\right)$ and $\cos\left(\frac{7\pi}{6}\right)$.
 $\frac{7\pi}{6} = 210^\circ$, Quadrant III, reference angle 30° . $\sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$, $\cos\left(\frac{7\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.
- 4 Find the exact value of $\tan\left(\frac{5\pi}{3}\right)$.
 $\frac{5\pi}{3} = 300^\circ$, Quadrant IV, reference angle 60° . $\sin = -\frac{\sqrt{3}}{2}$, $\cos = \frac{1}{2}$, so $\tan\left(\frac{5\pi}{3}\right) = \frac{-\sqrt{3}/2}{1/2} = -\sqrt{3}$.

Visualizing sine and cosine graphs

- 5 The graph shows $f(x) = \sin(x)$ and $g(x) = \cos(x)$ over one period.
 - a State the amplitude and period of each function shown. Both curves have amplitude 1 and period 2π , matching the single full up-down cycle shown across the interval.
 - b Identify the horizontal shift that maps $\sin(x)$ onto $\cos(x)$. $\cos(x) = \sin\left(x + \frac{\pi}{2}\right)$, so shifting $\sin(x)$ left by $\frac{\pi}{2}$ produces $\cos(x)$ — visible as the cosine curve reaching its peak a quarter-period earlier.

Amplitude, period, and vertical shift

- 6 State the amplitude, period, and vertical shift of $f(x) = 3\sin(2x) + 1$.
Amplitude = $|3| = 3$. Period = $\frac{2\pi}{2} = \pi$. Vertical shift = 1 (midline $y = 1$).
- 7 A Ferris wheel's height above the ground is modeled by $h(t) = 20\cos\left(\frac{\pi}{6}t\right) + 22$, where t is in minutes. Find the maximum height, minimum height, and period of one full rotation.
Maximum height = $22 + 20 = 42$ m. Minimum height = $22 - 20 = 2$ m. Period = $\frac{2\pi}{\pi/6} = 12$ minutes.

Reciprocal and Pythagorean identities

- 8 Given $\sin \theta = \frac{3}{5}$ with θ in Quadrant II, find $\cos \theta$ and $\tan \theta$.
 Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$, so $\cos \theta = \pm \frac{4}{5}$. In Quadrant II, cosine is negative:
 $\cos \theta = -\frac{4}{5}$. $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.
- 9 Find $\sec \theta$ and $\csc \theta$ if $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III.
 $\sec \theta = \frac{1}{\cos \theta} = -\frac{13}{5}$. In Quadrant III, $\sin \theta < 0$: using $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$, $\sin \theta = -\frac{12}{13}$, so
 $\csc \theta = -\frac{13}{12}$.

Modeling periodic phenomena

- 10 The average daily temperature in a city is modeled by $T(m) = 15\sin\left(\frac{\pi}{6}(m-3)\right) + 18$ degrees Celsius, where m is the month number ($m = 1$ for January). Find the temperature in April ($m = 4$) and state the month of maximum temperature.
 $T(4) = 15\sin\left(\frac{\pi}{6}(1)\right) + 18 = 15\sin\left(\frac{\pi}{6}\right) + 18 = 15(0.5) + 18 = 25.5^\circ \text{C}$. Maximum occurs when $\sin\left(\frac{\pi}{6}(m-3)\right) = 1$, i.e. $\frac{\pi}{6}(m-3) = \frac{\pi}{2}$, giving $m-3 = 3$, so $m = 6$ (June).

Co-function and even/odd identities

- 11 Use a co-function identity to rewrite $\cos\left(\frac{\pi}{5}\right)$ in terms of sine.
 $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, so $\cos\left(\frac{\pi}{5}\right) = \sin\left(\frac{\pi}{2} - \frac{\pi}{5}\right) = \sin\left(\frac{3\pi}{10}\right)$.
- 12 Use the fact that sine is an odd function and cosine is an even function to evaluate $\sin(-30^\circ)$ and $\cos(-60^\circ)$.
 Since $\sin(-\theta) = -\sin(\theta)$: $\sin(-30^\circ) = -\sin(30^\circ) = -\frac{1}{2}$. Since $\cos(-\theta) = \cos(\theta)$:
 $\cos(-60^\circ) = \cos(60^\circ) = \frac{1}{2}$.

Solving basic trigonometric equations

- 13 Solve $2\sin \theta - 1 = 0$ for θ in $[0, 2\pi)$.
 $\sin \theta = \frac{1}{2}$. Reference angle $\frac{\pi}{6}$, with sine positive in Quadrants I and II: $\theta = \frac{\pi}{6}$ or $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.
- 14 Solve $2\cos^2 \theta - 1 = 0$ for θ in $[0, 2\pi)$.
 $\cos^2 \theta = \frac{1}{2}$, so $\cos \theta = \pm \frac{1}{\sqrt{2}} = \pm \frac{\sqrt{2}}{2}$. Reference angle $\frac{\pi}{4}$, one solution in each quadrant:
 $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$.

Angular and linear speed

- 15** A wheel of radius 10 cm rotates at 4 revolutions per second. Find its angular speed ω in radians per second, and the linear speed of a point on its rim.
 $\omega = 4 \times 2\pi = 8\pi$ rad/s (4 revolutions per second, each revolution = 2π radians). Linear speed $v = r\omega = 10(8\pi) = 80\pi \approx 251.3$ cm/s.
- 16** A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its linear speed in km/h.
Angular speed $\omega = \frac{2\pi}{2} = \pi$ rad/h. Linear speed $v = r\omega = 7000\pi \approx 21,991$ km/h.
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Right-triangle definitions and applications

- 17** A ladder leans against a wall, making a 65° angle with the ground. If the ladder is 12 ft long, find the height it reaches on the wall, to one decimal place.
Height = $12\sin(65^\circ) \approx 12(0.9063) \approx 10.9$ ft.
- 18** From a point 50 ft from the base of a tower, the angle of elevation to the top is 40° . Find the height of the tower, to one decimal place.
Height = $50\tan(40^\circ) \approx 50(0.8391) \approx 42.0$ ft.
- 19** A kite string is 80 ft long and makes a 55° angle with the ground. Find the height of the kite above the ground, and the horizontal distance from the flyer to a point directly below the kite, both to one decimal place.
Height = $80\sin(55^\circ) \approx 80(0.8192) \approx 65.5$ ft. Horizontal distance = $80\cos(55^\circ) \approx 80(0.5736) \approx 45.9$ ft.